B. M. S. COLLEGE OF ENGINEERING, BANGALORE – 560019
DEPARTMENT OF MATHEMATICS

TRANSFORM CALCULUS, FOURIER SERIES AND NUMERICAL TECHNIQUES

(Course Code: 23MA3BSTFN)

UNIT-2: FOURIER SERIES

Dirichlet's conditions

Any function $f(x)$ can be developed as a ***FOURIER SERIES***

$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos nx+\sum_{n=1}^{\infty}b_n\sin nx$ where a_0,a_n,b_n are constants provided:

(i) $f(x)$ is periodic, single-valued and finite;

(ii) $f(x)$ has a finite number of discontinuities in any one period;

(iii) $f(x)$ has at the most a finite number of maxima and minima.

Fourier series expansion of a function $f(x)$ over $(a,a+2l)$ is

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos\left(\frac{n\pi}{l}x\right)+\sum_{n=1}^{\infty}b_n\sin\left(\frac{n\pi}{l}x\right),$$

Where $a_0=\frac{1}{l}\int_a^{a+2l}f(x)dx$, $\quad a_n=\frac{1}{l}\int_a^{a+2l}f(x)\cos\left(\frac{n\pi}{l}x\right)dx \quad n=1,2,3\ldots\ldots,$

and $b_n=\frac{1}{l}\int_a^{a+2l}f(x)\sin\left(\frac{n\pi}{l}x\right)dx \quad n=1,2,3\ldots\ldots.$

I. Fourier series expansion for the following periodic functions in the interval : $(0,2\pi)$

1. $f(x)=x$ **Ans:** $x=\pi-2\left(\sin x+\frac{\sin 2x}{2}+\ldots..\right)$.

2. $f(x)=x^2$ **Ans:** $f(x)=\frac{4\pi^2}{3}+\sum_{n=1}^{\infty}\left(\frac{4\cos nx}{n^2}-\frac{4\pi\sin nx}{n}\right)$.

3. $f(x)=\frac{\pi-x}{2}$ **Ans:** $\frac{\pi-x}{2}=\sum_{n=1}^{\infty}\frac{\sin nx}{n}$.

4. $f(x)=x(2\pi-x)$, **Ans:** $\frac{2\pi^2}{3}+4\sum_{i=1}^{\infty}\frac{cosnx-sinnx}{n^2}$

5. $f(x)=e^{-ax}$. **Ans:** $e^{-ax}=\frac{1-e^{-2a\pi}}{\pi}\left\{\frac{1}{2a}+\sum_{n=1}^{\infty}\frac{a\cos nx+n\sin nx}{n^2+a^2}\right\}$.

6. If $f(x)=\left[\frac{\pi-x}{2}\right]^2$, show that $f(x)=\frac{\pi^2}{12}+\sum_{n=1}^{\infty}\frac{\cos nx}{n^2}$.

Fourier series expansion for the following periodic functions over the interval $(-\pi,\pi)$:

1. $f(x)=x.$ $\quad Ans: x=2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx$.

2. $f(x)=|x|$, **Ans:** $|x|=\frac{\pi}{2}+\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n-1}{n^2}\cos nx$.

3. $f(x)=\pi^2-x^2$ **Ans:** $\pi^2-x^2=\frac{2\pi^2}{3}+4\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}\cos nx$.

4. $f(x)=x^2$ **Ans:** $x^2=\frac{\pi^2}{3}+4\sum_{n=1}^{\infty}\frac{(-1)^n\cos nx}{n^2}$.

5. $f(x)=\sin mx$ where m is neither zero nor an integer. **Ans:** $\sin mx=\sum_{n=1}^{\infty}\frac{2n(-1)^n\sin m\pi}{\left(m^2-n^2\right)\pi}\sin nx$.

6. $f(x)=\cos ax$ where 'a 'is not an integer. **Ans:** $\cos ax=\frac{2a\sin a\pi}{\pi}\left[\frac{1}{2a^2}+\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2-a^2}\cos nx\right]$.

7. $f(x)=x\cos x$ and deduce that $x\cos x=-\frac{1}{2}\sin x+\sum_{n=2}^{\infty}\frac{(-1)^n}{n^2-1}\sin nx$.

8. If $f(x)=x-x^2$ *and* $f(x)=x+x^2$

Ans:

$$x-x^2=-\frac{\pi^2}{3}+\sum_{n=1}^{\infty}(-1)^{n+1}\left\{\frac{4}{n^2}\cos nx+\frac{2}{n}\sin nx\right\}$$

$$x+x^2=\frac{\pi^2}{3}+\sum_{n=1}^{\infty}(-1)^{n}\left\{\frac{4}{n^2}\cos nx-\frac{2}{n}\sin nx\right\}$$

9. $f(x)=e^{-ax}$ **Ans:** $e^{-ax}=\frac{\sinh a\pi}{a\pi}+\frac{2\sinh a\pi}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{a^2+n^2}\left[a\cos nx+n\sin nx\right]$.

II. Fourier series expansion for the piece-wise continuous / discontinuous periodic functions:

1. $f(x)=\begin{cases}x-\pi/2 & -\pi<x<0\\ x+\pi/2 & 0<x<\pi\end{cases}$ **Ans:** $f(x)=\sum_{n=1}^{\infty}\frac{1-3(-1)^n}{n}\sin nx$.

2. $f(x)=\begin{cases}-\cos x & -\pi<x<0\\ \cos x & 0<x<\pi\end{cases}$. **Ans:** $f(x)=\frac{2}{\pi}-\frac{4}{\pi}\sum_{n=2}^{n}\frac{\cos\left(n\pi/2\right)}{n^2-1}\cos nx$.

3. $f(x)=\begin{cases}-1 & -\pi<x<0\\ 0 & x=0\\ 1 & 0<x<\pi\end{cases}$ **Ans:** $f(x)=\frac{4}{\pi}\left[\sin x+\frac{\sin 3x}{3}+\frac{\sin 5x}{5}.....\infty\right]$.

4. $f(x)=\begin{cases}-\pi & -\pi<x<0\\ x & 0<x<\pi\end{cases}$ and hence deduce that sum of the reciprocal squares of odd integers is equal to $\frac{\pi^2}{8}$. **Ans:** $f(x)=\frac{-\pi}{4}-\frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\left[1-(-1)^n\right]}{n^2}\cos nx+\sum_{n=1}^{\infty}\left[\frac{1-2(-1)^n}{n}\right]\sin nx$.

5. $f(x)=\begin{cases}0 & -\pi\le x\le 0\\ x & 0<x\le\pi\end{cases}$.**Ans:** $f(x)=\frac{\pi}{4}-\frac{2}{\pi}\left[\frac{\cos x}{1^2}+\frac{\cos 3x}{3^2}+\cdots\cdots\right]+\left[\frac{\sin x}{1}-\frac{\sin 2x}{2}+\cdots\cdots\right]$

6. $f(x)=\begin{cases}-k & -\pi<x<0\\ k & 0<x<\pi\end{cases}$ **Ans:** $f(x)=\frac{4k}{\pi}\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{2n-1}$.

7. $f(x)=\begin{cases}-(\pi+x)/2 & -\pi\le x<0\\ (\pi-x)/2 & 0\le x<\pi\end{cases}$. **Ans:** $f(x)=\sin x+\frac{\sin 2x}{2}+\frac{\sin 3x}{3}+\cdots\cdots$.

8. $f(x)=\begin{cases}-x^2 & -\pi\le x<0\\ x^2 & 0\le x<\pi\end{cases}$. **Ans:** $f(x)=\frac{2}{\pi}\sum_{n=1}^{\infty}\left[\frac{\pi^2}{n}(-1)^{n+1}+\frac{2}{n^3}\left\{1-(-1)^n\right\}\right]\sin nx$.

9. $f(x)=\begin{cases}1+2x/\pi & -\pi<x\le 0\\ 1-2x/\pi & 0\le x<\pi\end{cases}$ **Ans:** $f(x)=\frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^2}\cos nx.$.

10. $f(x)=\begin{cases}\pi-x & 0\le x\le\pi\\ x-\pi & \pi\le x\le 2\pi\end{cases}$. **Ans:** $f(x)=\frac{\pi}{2}+\frac{4}{\pi}\sum_{n=1,3,5.....}\frac{1}{n^2}\cos nx..$

11. $f(x)=\begin{cases}-x & -\pi<x\le 0\\ x & 0\le x<\pi\end{cases}$ **Ans:** $f(x)=\frac{\pi}{2}-\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^2}\cos nx$.

13. $f(x)=\begin{cases}0 & -\pi\le x\le 0\\ x^2 & 0\le x\le\pi\end{cases}$ which is assumed to be periodic with period 2π.

Ans: $f(x)=\frac{\pi^2}{6}+2\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx+\frac{1}{\pi}\sum_{n=1}^{\infty}(-1)^n(\frac{2}{n^3}-\frac{\pi^2}{n})\sin nx$.

14. $f(x)=\begin{cases}-x+1 & -\pi\le x\le 0\\ x+1 & 0\le x\le\pi\end{cases}$. Is the function even or odd?

Ans: $f(x)=\frac{\pi}{2}+1-\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2}$.

15. Show that the Fourier series of $f(x)=\begin{cases}x & 0\le x\le\pi\\ 2\pi-x & \pi\le x\le 2\pi\end{cases}$ is

$f(x)=\frac{\pi}{2}-\frac{4}{\pi}\left[\frac{\cos x}{1^2}+\frac{\cos 3x}{3^2}+\cdots\cdots\right]$.

III. <u>Fourier series expansion for arbitrary period :</u>

1. $f(x)=|x|$. **Ans:** $f(x)=\frac{l}{2}-\frac{2l}{\pi^2}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^2}\cos\frac{n\pi}{l}x$.

2. $f(x)=e^x,\quad(-l,l)$. **Ans:** $f(x)=\sinh l\left[\frac{1}{l}+2l\sum_{n=1}^{\infty}(-1)^n\frac{\cos n\pi x/l}{l^2+n^2\pi^2}+2\pi\sum_{n=1}^{\infty}(-1)^{n-1}\frac{n\sin n\pi x/l}{l^2+n^2\pi^2}\right]$

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3. $f(x)=x(2l-x),\ (0,2l)$ **Ans:** $\frac{2l^2}{3}+\frac{4l^2}{\pi^2}\sum_{i=1}^{\infty}\frac{(\cos\frac{n\pi x}{l}-\sin\frac{n\pi x}{l})}{n^2}$

4. $f(x)=\sin mx\ in\ (-l,l)$. **Ans:** $f(x)=\sin ml\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2\pi^2-m^2l^2}\sin\frac{n\pi x}{l}$.

5. $f(x)=x+x^2\quad(-1,1)$. **Ans:** $f(x)=\frac{1}{3}+\frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos n\pi x+\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin n\pi x$.

6. $f(x)=x^2\quad(-l,l)$. **Ans:** $f(x)=\frac{l^3}{3}+\sum_{n=1}^{\infty}\frac{(-1)^n4l^2}{n^2\pi^2}\cos\frac{n\pi x}{l}$.

7. $f(x)=e^{-x}$ in the interval $0<x<2$.

Ans: $f(x)=\frac{e^2-1}{2e^2}+\sum_{n=1}^{\infty}\frac{e^2-1}{e^2(1+n^2\pi^2)}\cos n\pi x+\sum_{n=}^{\infty}\frac{n\pi(e^2-1)}{e^2(1+n^2\pi^2)}\sin n\pi x$.

8. $f(t)=4-t^2\ in\ (-2,2)$. **Ans:** $f(x)=\frac{8}{3}+\frac{16}{\pi^2}\sum\frac{(-1)^{n+1}}{n^2}\cos\frac{n\pi t}{2}$.

9. $f(x)=2x-x^2\quad(0,3)$ **Ans:** $f(x)=\sum_{n=1}^{\infty}\frac{9}{n^2\pi^2}\cos\left(\frac{2n\pi x}{3}\right)+\sum_{n=1}^{\infty}\frac{3}{n\pi}\sin\left(\frac{2n\pi x}{3}\right)$.

10. $f(x)=\pi x\quad(0,2)$. **Ans:** $f(x)=-\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin n\pi x}{n}$.

11. $f(x)=\frac{\pi-x}{2}\ in\ 0<x<2$. **Ans:** $\frac{\pi-x}{2}=\frac{\pi-1}{2}+\sum_{n=1}^{\infty}\frac{\sin n\pi x}{n\pi}$.

12. $f(x)=\begin{cases}1+\dfrac{4x}{3} & -\dfrac{3}{2}<x<0\\ 1-\dfrac{4x}{3} & 0<x<\dfrac{3}{2}\end{cases}$

13. $f(x)=\begin{cases}\pi x & 0\le x\le 1\\ \pi(2-x) & 1\le x\le 2\end{cases}$ show that Fourier expansion of the function

$f(x)=\dfrac{\pi}{2}-\dfrac{4}{\pi}\left[\dfrac{\cos\pi x}{1^2}+\dfrac{\cos 3\pi x}{3^2}+\dfrac{\cos 5\pi x}{5^2}+\dots\dots\dots\dots\right]$ and deduce that $\dfrac{\pi^2}{8}=\sum_{n=1}^{\infty}\dfrac{1}{(2n-1)^2}$.

14. If $f(x)=\begin{cases}2-x & in \quad 0\le x\le 4\\ x-6 & in \quad 4\le x\le 8\end{cases}$, express $f(x)$ as a Fourier series and hence deduce that $\dfrac{\pi^2}{8}=\sum_{n=1}^{\infty}\dfrac{1}{(2n-1)^2}$. **Ans:** $f(x)=\dfrac{8}{\pi^2}\sum_{n=1}^{\infty}\dfrac{1-(-1)^n}{n^2}\cos\dfrac{n\pi}{4}x$.

15. $f(x)=\begin{cases}0 & -2<x\le -1\\ 1+x & -1<x<0\\ 1-x & 0\le x<1\\ 0 & 1\le x<2\end{cases}$. **Ans:** $f(x)=\dfrac{1}{4}+\dfrac{4}{\pi^2}\sum_{n=1}^{\infty}\dfrac{1}{n^2}\left[1-\cos\dfrac{n\pi}{2}\right]\cos\dfrac{n\pi x}{2}$.

16. $f(x)=\begin{cases}x & -1<x\le 0\\ x+2 & 0<x\le 1\end{cases}$ **Ans:** $f(x)=1+\dfrac{2}{\pi}\sum_{n=1}^{\infty}\dfrac{1}{n}[1-2(-1)^n]\sin n\pi x$.

17. $f(x)=\begin{cases}l-x & 0<x\le l\\ 0 & l\le x\le 2l\end{cases}$

18. $f(x)=\begin{cases}\dfrac{l}{2}-x & (-l,0)\\ \dfrac{l}{2}+x & (0,l)\end{cases}$ **Ans:** $f(x)=l-\dfrac{4l}{\pi^2}\sum_{n=1,3,5....}^{\infty}\dfrac{1}{n^2}\cos\left(\dfrac{n\pi x}{l}\right)$.

19. $f(x)=\begin{cases}x & 0<x<1\\ 0 & 1<x<2\end{cases}$ **Ans:** $f(x)=\dfrac{1}{4}+\dfrac{1}{\pi^2}\sum_{n=1}^{\infty}\dfrac{(-1)^n-1}{n^2}\cos n\pi x+\dfrac{1}{\pi}\sum_{n=1}^{\infty}\dfrac{(-1)^{n-1}\sin n\pi x}{n}$.

<u>Complex Fourier series</u>

$f(x)=\sum_{n=-\infty}^{\infty}c_n e^{in\pi x/L}$ where $c_n=\dfrac{1}{2L}\int_{-L}^{L}f(x)e^{-in\pi x/L}dx$

Find the complex form of the Fourier series of the periodic function

1. $f(x)=e^x$ in $-\pi<x<\pi$
2. $f(x)=e^{-x}$ in $-1\le x\le 1$. **Ans:** $f(x)=\sum_{n=-\infty}^{\infty}\dfrac{(-1)^n(1-in\pi)}{1+n^2\pi^2}\sinh 1 e^{in\pi x}$
3. $f(x)=\sin t$ in $0<t<\pi$. **Ans:** $f(x)=\dfrac{1}{\pi}\left[\dfrac{1}{2}-\dfrac{\cosh 2t}{1.3}-\dfrac{\cosh 4t}{3.5}-\dfrac{\cosh 6t}{5.7}-.....\right]$
4. $f(x)=\cos ax$ in $-\pi<x<\pi$. **Ans:** $f(x)=\dfrac{a}{\pi}\sin a\pi\sum_{n=-\infty}^{\infty}\dfrac{(-1)^n e^{inx}}{a^2-n^2}$
5. $f(x)=\begin{cases}0 & in \quad 0<x<l\\ a & in \quad l<x<2l\end{cases}$.

Ans: $f(x)=\frac{a}{2}-\frac{2a}{\pi}\left[\sinh u+\frac{\sinh 3u}{3}+\frac{\sinh 5u}{5}+.....\right]$ where $u=\frac{i\pi x}{l}$

6. $f(x)=x \; in \; -\pi<x<\pi$.
7. $f(x)=x^2 \; in \; -l<x<l$.
8. $f(x)=\begin{cases}-k & in \quad -\pi<x<0\\ k & in \quad 0<x<\pi\end{cases}$.

Harmonic Analysis

Fourier series expansion of $f(x)$ over $(a,a+2l)$ is

$$f(x)=\frac{a_0}{2}+\left(a_1\cos\frac{\pi}{l}x+b_1\sin\frac{\pi}{l}x\right)+\left(a_2\cos\frac{2\pi}{l}x+b_2\sin\frac{2\pi}{l}x\right)+----$$

Where $a_0=2\left[\text{Mean value of } f(x) \text{ in} (a,a+2l)\right]$

$$a_n=2\left[\text{Mean value of } f(x)\cos\frac{n\pi}{l}x \text{ in } (a,a+2l)\right];$$

$$b_n=2\left[\text{Mean value of } f(x)\sin\frac{n\pi}{l}x \text{ in } (a,a+2l)\right] \quad n=1,2\cdots\cdots$$

1. Express y as a Fourier series up to the second harmonics given

θ:	0	60	120	180	240	300	360
y:	4	3	2	4	5	6	4

. **Ans:** $y=4+(0.33\cos x-1.732\sin x)$

2. The following data gives the variations of a periodic current over a period

t sec :	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
i amp:	1.98	1.3	1.05	1.3	-0.88	-0.25	1.98

Show that there is a direct current part of 0.75 amp in the variable current and obtain the amplitude of the first harmonic. **Ans:** $a_0=0.75$ $a_1=0.3733$ $b_1=1.00456$ amplitude=1.0717

3. Express y as a Fourier series up to the second harmonics given

x:	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y:	1.98	1.30	1.05	1.3	-0.88	-0.25	1.98

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Ans: $y=0.75+(0.3733\,cos\,x+1.00456\,sin\,x)+10.89\,cos\,2x-0.1097\,sin\,2x)$

4. Obtain the constant term and the co-efficient of $\sin\theta$ and $\sin 2\theta$ in the Fourier series expansion of y from the following data.

θ:	0	60	120	180	240	300	360
y:	0	9.2	14.4	17.8	17.3	11.7	0

. **Ans:** $a_0=23.466$ $b_1=0$ $b_2=-1.566$

5. Obtain the constant term and the first two harmonic term of the Fourier series expansion of y from the following data.

x:	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y:	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Ans: a_0=2.9, $(-0.367\cos x+0.1732\sin x)$ and $(-0.1\cos 2x-0.0577\sin 2x)$.

6. Express y as a Fourier series up to the second harmonics given

x:	0	1	2	3	4	5
y:	4	8	15	7	6	2

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Ans: $y=7-2.833\,cos\frac{\pi x}{3}-1.5\,cos\frac{2\pi x}{3}+4.33\,sin\frac{\pi x}{3}-0.866\,sin\frac{2\pi x}{3}$

7. Obtain the constant term and the first harmonic term of the Fourier series expansion of y from the following data.

x:	0	1	2	3	4	5
y:	9	18	24	28	26	20

Ans: $a_0=41.6666$ $a_1=-8.3334$ $b_1=-1.1548$

8. In a machine the displacement y of a given point is given for certain angles of θ

θ: 0 30 60 90 120 150 180 210 240 270 300 330

y: 7.9 8 7.2 5.6 3.6 1.7 0.5 0.2 0.9 2.5 4.7 6.8.

Find the co-efficient of $\sin 2\theta$ in the Fourier expansion of y. **Ans:** $a_2 = -0.0731$

9. Given the following table

θ:	0	60	120	180	240	300	360
y:	7.9	7.2	3.6	0.5	0.9	6.8	7.9

. Obtain the Fourier series neglecting terms higher than first harmonics. **Ans:** $f(x) = 4.4833 + 4.05\cos x + 0.8949\sin x$.

10. The turning moment T on the crank shaft of a steam engine for the crank angle θ is given as follows:

θ: 0 30 60 90 120 150 180 210 240 270 300 330

y: 0 2.7 5.2 7 8.1 8.3 7.9 6.8 5.5 4.1 2.6 1.2

Expand T as a Fourier series upto first harmonics. **Ans:** $T = 4.95 - 3.4165\cos\theta + 1.4839\sin\theta$

11. The displacement y of a part of a mechanism is tabulated with corresponding angular movement x^0 of the crank. Express y as a Fourier series neglecting the harmonics above the third.

θ	30	60	90	120	150	180	210	240	270	300	330	360
y	1.8	1.1	0.3	0.16	1.5	1.3	2.16	1.25	1.30	1.52	1.76	2

Ans: $y = 1.2625 + (0.0414\cos x - 0.6252\sin x) + (0.5292\cos 2x - 0.2335\sin 2x)$

12. Express y as a Fourier series up to the second harmonics given

θ: 30 60 90 120 150 180 210 240 270 300 330 360

y: 2.34 3.01 3.68 4.15 3.69 2.20 0.83 0.51 0.88 1.09 1.19 1.64.

Ans: $y = 2.102 + (0.558\cos x + 1.531\sin x) + (0.354\cos 2x + 0.145\sin 2x)$.

13. The following data gives the variations of a periodic current over a period.

x:	0	2	4	6	8	10	12
y:	9.0	18.2	24.4	27.8	27.5	22.0	9.0

. Find the Fourier expansion up to the second harmonic. **Ans:** $y = 21.4835 + (-8.217\cos\theta - 1.9918\sin\theta) + (-3.083\cos 2\theta - 0.202\sin 2\theta)$